PRODUCT REQUIREMENT DOCUMENT

PRD — Laundry POS & Management System

PRD-2026 · AIAvenue Agency

Executive Summary

India's laundry and dry-cleaning market is a ₹12,000 Cr industry, with over 80% comprising unorganised neighbourhood dhobi shops and small-format laundromats. These businesses operate on paper-based billing, manual inventory tracking, and WhatsApp-based order management — resulting in 15–22% revenue leakage through lost orders, incorrect pricing, and untracked inventory. This product is a mobile-first, dark-themed POS and management system tailored for Indian laundry businesses. It packages 19 distinct features — including a 4-step order wizard, AI Morning Briefing, Agentic Chat POS powered by Claude `tool_use`, Claude Vision garment detection, Smart Pricing, Sentiment Analysis, Re-engagement Writer, Analytics Narrative, CRM with segmentation, anomaly detection, inventory management, PDF receipts, QR codes, Cmd+K palette, notification drawer, and collapsible sidebar — into a single React 18 application with Framer Motion animations. Target TAM is ₹360 Cr, SAM ₹72 Cr, SOM ₹14 Cr in the Indian laundry-tech SaaS market.

Problem Statement

Core Problem

Indian laundry businesses are stuck in a low-tech equilibrium. The average neighbourhood laundry processes 60–120 garments daily but tracks everything in a ruled notebook. The shift to organised laundry services (UClean, LaundroKart, FabCare) has raised customer expectations, yet the backend operations of most independent stores have not evolved.

User Pain Points

Persona	Pain Point	Frequency	Severity
Store Owner	No visibility into daily P&L; doesn't know which service is most profitable	Daily	Critical
Billing Clerk	1–2 minutes per order on paper; errors in pricing (pricing has 6+ variables: fabric, item type, service, add-ons, weight, discount)	Per order	High
Customer	No digital receipt; no order tracking; no re-order convenience	Per visit	Medium
Franchise Operator	Cannot compare performance across 5+ stores without manual Excel consolidation	Weekly	High
Delivery Boy	Picks up wrong items; no photo evidence of garment condition at pickup	Per pickup	Medium

Core Problem

Indian laundry businesses are stuck in a low-tech equilibrium. The average neighbourhood laundry processes 60–120 garments daily but tracks everything in a ruled notebook. The shift to organised laundry services (UClean, LaundroKart, FabCare) has raised customer expectations, yet the backend operations of most independent stores have not evolved.

User Pain Points

Revenue Leakage Sources

- **15–22%** leakage from unrecorded orders, incorrect pricing, untracked free services
- **8–10%** of orders are disputed due to missing garment-condition documentation
- **12%** of customers never return — no re-engagement system exists
- Manual inventory tracking leads to **₹3,000–₹8,000/month** in lost or unaccounted inventory consumables

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User Research & Personas

Persona 1: Prakash Joshi — Store Owner

Attribute	Detail
Age	38
Location	Pune — Kothrud
Business	2 laundry stores; 12 employees; 180 garments/day
Tech Comfort	Moderate; uses WhatsApp Business, Google Pay, Paytm
Workflow	Wakes at 6 AM, reviews yesterday's notebook, gives cash to clerk, reconciles at 10 PM
Frustration	"I don't know if I made profit or loss until I sit at night with a calculator. By then, the day's mistakes are done."
Need	Real-time P&L dashboard, automated billing, inventory alerts, staff performance metrics

Persona 1: Prakash Joshi — Store Owner

Persona 2: Sunita Verma — Billing Clerk

Persona 3: Rohan Deshmukh — Franchise Operator (4 stores)

Market Analysis

Total Addressable Market (TAM)

Indian laundry services market (organised + unorganised): **₹12,000 Cr** (2026, CAGR 8%).

Technology spend as a percentage of revenue in small retail businesses: **~3%** (conservative).

TAM = ₹12,000 Cr × 3% = ₹360 Cr

Serviceable Addressable Market (SAM)

Focus on the **organised semi-organised segment**: standalone laundry shops with ≥2 employees and ≥50 garments/day throughput that have adopted at least one digital payment method. Estimated **20%** of the total market.

SAM = ₹360 Cr × 20% = ₹72 Cr

Serviceable Obtainable Market (SOM)

Target 2,000 stores in Tier-1 and Tier-2 cities over 3 years at an average ARPU of ₹7,000/year (₹583/month).

SOM = 2,000 × ₹7,000 = ₹1.4 Cr/year (Year 3 target) → scaled to ₹14 Cr over 5 years

Market Rationale

- Only 8% of Indian laundry businesses use any POS software (vs. 65% in restaurants)
- The organised laundry segment is growing at 25% CAGR; unorganised is converting
- UPI adoption in small retail crossed 92% — digital readiness is no longer a barrier
- No dominant POS player exists for laundry-specific workflows (most use generic retail POS)

Competitive Analysis

Competitor	Type	Strengths	Weaknesses	Our Edge
Gofrugal	Retail POS	Strong distribution; inventory management	Generic retail; no laundry-specific pricing matrix; no garment tracking	Laundry-native features: fabric-based pricing, garment tagging, Claude Vision damage detection
dot Laundry POS	Laundry-specific	Basic order management; laundry workflow	Outdated UI (no dark theme); no AI; no CRM; poor mobile UX	Modern dark-theme UI; AI-powered features (Vision, Pricing, Sentiment, Briefing)
KhataBook	Ledger app	Massive India adoption; free; easy	No POS; no inventory; no billing — just ledger	Full-featured POS + ledger + CRM in one
Zoho Inventory	Inventory management	Enterprise-grade; multi-channel	Too complex for a 2-person laundry shop; no laundry domain features	Simplicity-first design for small laundry biz
Manual (Notebook + WhatsApp)	None	Zero cost; familiar	22% revenue leakage; no data; no growth	Clear ROI case — ₹2,000/month for 22% leakage recovery

Key Differentiators:

- 4-step order wizard optimised for speed (customer → items → services → payment; <30 seconds)
- Claude Vision garment detection (photo → auto-identify item type, fabric, colour)
- Claude `tool_use` Agentic Chat POS — natural language order entry
- Smart Pricing Engine (item + fabric + service + weight + add-ons + seasonal dynamic pricing)
- AI Morning Briefing (daily predicted revenue, yesterday's anomaly, staff alerts)
- Sentiment Analysis on customer feedback (gauge NPS from casual interactions)
- Re-engagement Writer (AI drafts SMS/WhatsApp messages to dormant customers)
- Analytics Narrative (plain-English daily/weekly report, not just charts)

- Framer Motion micro-interactions throughout for delightful UX

Solution Overview

High-Level Architecture

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│ Frontend (React 18) │
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│ │ POS │ │ Dashboard│ │ CRM │ │ Settings / │ │ │ │ (4-step │ │ (Morning │ │ (Segments,│ │ Inventory / │ │ │ │ Wizard) │ │ Briefing)│ │ History) │ │ Staff Management │ │
│ └──────────┘ └──────────┘ └──────────┘ └──────────────────┘ │
│ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────────────┐ │
│ │ Chat POS │ │ Analytics│ │ Notification│ Collapsible │ │ │ │ (Claude │ │ Narrative│ │ Drawer │ Sidebar + Cmd+K │ │ │ tool_use)│ │ │ │ │ │ │ │ │
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│ API Layer (Node.js + Express / Firestore + REST) │
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│ │ Claude API (Vision + tool_use + completions) │
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│ │ PDF Generator (jsPDF) / QR Code (qrcode) │
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│ │ Firestore / PostgreSQL (Orders, Customers, Inventory) │
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19 Features Map

#	Feature	Category	AI-Powered
1	4-Step Order Wizard (Customer → Items → Services → Payment)	Core POS	No
2	AI Morning Briefing (daily revenue prediction + alerts)	Dashboard	Yes (Claude)
3	Agentic Chat POS (Claude `tool_use` — "Add 3 shirts, 2 trousers for Sharma ji")	POS	Yes (Claude)
4	Claude Vision Garment Detection (photo → auto-identify item)	POS	Yes (Claude Vision)
5	Smart Pricing Engine (item + fabric + service + weight + seasonal)	Pricing	Yes (Rules + ML)
6	Sentiment Analysis (customer feedback → happy/neutral/angry)	CRM	Yes (Claude)

7	Re-engagement Writer (AI drafts SMS for dormant customers)	CRM	Yes (Claude)
8	Analytics Narrative (plain-English daily "story" of the business)	Analytics	Yes (Claude)
9	CRM with Segmentation (by frequency, avg spend, last visit)	CRM	No
10	Anomaly Detection (unusual order patterns, pricing errors)	Analytics	Yes (ML)
11	Inventory Management (detergent, packaging, consumables)	Operations	No
12	PDF Receipts (downloadable, printable, WhatsApp-shareable)	POS	No
13	QR Codes (order tracking link, digital business card)	Marketing	No
14	Cmd+K Palette (quick actions — search, navigate, create order)	UX	No
15	Notification Drawer (order status, payment alerts, re-order reminders)	UX	No
16	Collapsible Sidebar (responsive navigation)	UX	No
17	Dark Theme (default, battery-friendly for all-day use)	UX	No
18	Framer Motion Animations (transitions, micro-interactions)	UX	No
19	Staff Management (2 role levels: Clerk, Manager + permissions)	Operations	No

High-Level Architecture

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│ │ Chat POS │ │ Analytics│ │ Notification│ Collapsible │
│ │ (Claude │ │ Narrative│ │ Drawer │ Sidebar + Cmd+K │
│ │ tool_use)│ │ │ │
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19 Features Map

Feature Deep-Dive — RICE Priority Matrix

Feature	Reach (users/month)	Impact (1–5)	Confidence (%)	Effort (weeks)	RICE Score
4-Step Order Wizard	150	5.0	95%	4	178.1
Smart Pricing Engine	150	4.5	80%	3	180.0
PDF Receipts + QR Codes	150	4.0	90%	2	270.0
Dark Theme + Framer Motion	150	3.5	95%	3	166.3
Inventory Management	50	4.0	85%	4	42.5
AI Morning Briefing	50	4.5	70%	4	39.4
Agentic Chat POS (Claude)	80	4.5	65%	6	39.0
Claude Vision Garment Detection	80	4.0	60%	5	38.4
Analytics Narrative	30	4.0	70%	3	28.0
CRM with Segmentation	30	3.5	80%	3	28.0
Collapsible Sidebar + Cmd+K	150	3.0	90%	2	202.5
Notification Drawer	150	3.0	85%	2	191.3
Sentiment Analysis	30	3.0	55%	3	16.5
Re-engagement Writer	30	3.5	50%	2	26.3
Anomaly Detection	30	3.5	45%	4	11.8

Staff Management	50	3.0	90%	2	67.5

Priority Ranking (Tiered)

Tier 1 — Must-Have MVP (Weeks 1–6): 1. PDF Receipts + QR Codes (270.0) — Trivial effort, huge value 2. Collapsible Sidebar + Cmd+K (202.5) — UX foundation 3. Notification Drawer (191.3) — Essential UX 4. Smart Pricing Engine (180.0) — Core value prop 5. 4-Step Order Wizard (178.1) — Primary workflow 6. Dark Theme + Framer Motion (166.3) — Visual identity

Tier 2 — Growth (Weeks 7–12): 7. Staff Management (67.5) 8. Inventory Management (42.5) 9. AI Morning Briefing (39.4) 10. Agentic Chat POS (39.0) 11. Claude Vision Detection (38.4) 12. CRM with Segmentation (28.0) 13. Analytics Narrative (28.0)

Tier 3 — Differentiation (Weeks 13–16): 14. Re-engagement Writer (26.3) 15. Sentiment Analysis (16.5) 16. Anomaly Detection (11.8)

Success Metrics

KPI	Baseline	Target (3 months, 50 stores)	Target (12 months, 500 stores)
Monthly Active Stores	0	50	500
Orders/month/store	0 (new)	900	1,500
Average order time (clerk)	90 seconds (paper)	25 seconds	15 seconds
Revenue leakage recovered	22% (baseline)	≤ 8%	≤ 3%
Customer re-order rate (within 30 days)	~15% (no system)	30%	45%
Clerk onboarding time	N/A	2 hours	30 minutes
Store NPS (owner survey)	N/A	≥ 50	≥ 70
AI feature adoption (% of stores using Chat POS)	0%	20%	50%

Roadmap

Phase 1 — Core POS (Weeks 1–6)

Feature	Detail
4-Step Order Wizard	Customer selection → item entry → services → payment
Smart Pricing Engine	Item type + fabric + service + weight → auto price
PDF Receipts	Download, print, WhatsApp share with QR tracking link
Dark Theme + Framer Motion	Complete visual identity and micro-interactions
Collapsible Sidebar + Cmd+K	Navigation and quick-action palette

Phase 1 — Core POS (Weeks 1–6)

Go/No-Go: <30 second order time; 95% pricing accuracy on test dataset

Phase 2 — AI & Management (Weeks 7–12)

Go/No-Go: Chat POS accuracy ≥ 85%; Vision detection accuracy ≥ 80%

Phase 3 — Intelligence & Growth (Weeks 13–16)

Go/No-Go: Store retention ≥ 80% after 3 months; ARPU ≥ ₹583

Impact Estimate

Quantitative Impact (Annualized at 500 Stores)

Metric	Per Store (Annual)	500 Stores Total
Revenue leakage recovered (22% → 3%)	₹1,14,000 saved	₹5.7 Cr
New customer re-order rate improvement (15% → 45%)	₹72,000 incremental	₹3.6 Cr
Clerk productivity (90s → 15s per order = 150 hrs/year saved)	₹18,750 saved	₹0.94 Cr
Inventory waste reduction (₹6,000 → ₹1,000/month)	₹60,000 saved	₹3.0 Cr
Total Gross Value Created	**₹2,64,750/store**	**₹13.24 Cr**

Quantitative Impact (Annualized at 500 Stores)

SaaS Revenue Model

Qualitative Impact

- **Formalisation**: Brings unorganised laundry into the digital economy
- **Customer trust**: Digital receipts, order tracking, garment photo proof
- **Scalability**: Store owner can open 3rd, 4th, 5th location without proportional overhead
- **Employment**: Upskill clerks from notebook-entry to digital POS operation

Break-Even

- Development cost (3-person team for 4 months): **₹24 Lakh**
- Monthly infra (Firestore, Claude API, hosting): **₹1.5 Lakh/month**
- Customer acquisition cost: **₹2,000/store** (first 500 → ₹10 Lakh)
- **Break-even at ~280 stores** (≈14 months post-launch)
- **Year 3 ARR target: ₹1.2 Cr**

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